

A skin-conformal rigid-in-soft array-based imaging system

Xiaoyun Yuan^{1,†,‡}, Jianing Zhang^{1,†}, Youhai Liu¹, Yichen Wang¹, Lu Fang^{1,2,*}

¹ Department of Electronic Engineering, Beijing National Research Center for Information Science and Technology (BNRist), Tsinghua University, China.

² Institute for Brain and Cognitive Science, Tsinghua University (THUIBCS), Beijing, China.

[†] These authors contributed equally to this work.

[‡] Present address: MoE Key Lab of Artificial Intelligence, AI Institute, School of Computer Science, Shanghai Jiao Tong University, Shanghai, China

* Corresponding author. Email: fanglu@tsinghua.edu.cn.

Abstract

Array-based imaging systems offer unparalleled capabilities for mapping complex environments, but their inherent structural rigidity prevents intimate integration with dynamic, non-rigid hosts such as human skin and soft robots. While flexible visual sensor arrays offer compliance, fabrication constraints severely limit their pixel density and overall resolution. Here, we present SkinSight, a skin-conformal imaging system that overcomes this trade-off through a hybrid “rigid-in-soft” architecture. By integrating miniaturized imagers with stretchable interconnects, SkinSight achieves stable, high-resolution imaging under complex mechanical deformations. Furthermore, to resolve processing challenges induced by continuous deformation, we leverage large-scale pre-trained vision transformers for end-to-end geometric inference. This deformation-adaptive framework no longer relies on rigid structural priors, enabling robust, calibration-free 3D visual perception. We demonstrate seamless integration with human skin, transforming a human palm into a sensory interface capable of perceiving objects beyond the direct line of sight. By augmenting the perceptual capabilities of humans and robots, SkinSight demonstrates broad potential for next-generation wearables and embodied artificial intelligence.

Introduction

Array-based imaging systems excel at mapping complex environments, making them essential for modern platforms like smartphones and autonomous vehicles¹⁻⁵. Their exceptional performance fundamentally relies on a spatially distributed architecture, which establishes the large effective aperture and extended baseline required for advanced visual tasks. However, maintaining this spatial distribution demands structural rigidity, thereby preventing integration with dynamic, non-rigid hosts such as humans and soft robots. While individual camera modules can undergo miniaturization, the overall array cannot. Shrinking the overall footprint inherently collapses the system into a single viewpoint, eliminating the core advantages of multi-aperture sensing. Consequently, the requirement to maintain a large physical span makes structural rigidity a fundamental barrier to their intimate integration with deformable hosts.

Driven by the need for conformal integration, recent research has explored intrinsically flexible vision sensors based on advanced fabrication techniques or novel materials, including serpentine⁶ or compressible copper wires⁷⁻⁹, nanowire composites^{10,11}, kirigami¹² or origami-inspired¹³ silicon structures, and materials like quantum dots and perovskites¹⁴⁻¹⁷. However, these systems face severe fabrication bottlenecks. Mechanical and electrical limitations restrict the ability to shrink individual pixel sizes while simultaneously scaling up the total pixel count. Consequently, these arrays suffer from low pixel counts, rendering them inadequate for advanced visual tasks that typically demand megapixel-level fidelity. Conversely, while mature semiconductor technologies yield densely integrated and high-performance arrays, fabrication constraints inherently confine them to planar inflexible platforms. While unstructured camera arrays have been developed to introduce internal spatial adaptability, their design purpose differs fundamentally from conforming to dynamic hosts^{2-5,18}. These systems merely reconfigure camera orientations within a rigid supporting body. This continued reliance on a rigid backbone prevents their intimate integration with non-rigid surfaces.

To overcome these limitations, we present SkinSight, a skin-conformal imaging system built upon a hybrid “rigid-in-soft” architecture. This design physically decouples optical sensing from mechanical strain by utilizing rigid components to protect high-density sensor arrays and a soft matrix to absorb overall deformation. A key advantage of this format is its ability to provide high-

57 resolution vision without altering the host’s internal structure or compromising its inherent
58 mechanical properties, thereby enabling a non-invasive sensory augmentation that is highly
59 adaptable to diverse dynamic surfaces, from human skin to soft robotic platforms. Specifically, we
60 integrate miniaturized commercial CMOS imagers as high-fidelity rigid nodes onto a flexible
61 substrate. To stably transmit high-throughput image signals under continuous motion, we employ
62 stretchable serpentine interconnects designed to accommodate complex deformations, such as
63 bending, twisting, and stretching. Fundamentally, SkinSight functions as a skin-conformal
64 unstructured camera array. By liberating this array architecture from traditional rigid backbones,
65 our system effectively bridges the critical gap between high-resolution machine vision and
66 mechanically compliant bio-integrated electronics.

67 While this rigid-in-soft architecture solves the physical integration problem, it introduces a
68 major computational challenge. As the array conforms to dynamic hosts, the relative poses of
69 individual cameras undergo continuous variations. To address this issue, we leverage recent
70 breakthroughs in large-scale pretrained vision transformers to develop a deformation-adaptive
71 computational framework. This approach enables the fusion of multi-perspective information
72 without explicit spatial calibration, effectively liberating array-based imaging from the
73 requirement of rigid structural priors. By enabling robust 3D visual perception from continuously
74 varying inputs, SkinSight overcomes the fundamental limitations of static imaging geometries.

75 We demonstrate the practical utility of SkinSight through its seamless integration with
76 complex dynamic hosts, including human skin and soft robots. In particular, attaching the system
77 to the human palm transforms the hand into an active sensory interface, empowering users to
78 perceive and reconstruct objects beyond their direct line of sight. By evolving array-based imaging
79 from rigid platforms to flexible skin-conformal formats through AI-driven data fusion, SkinSight
80 demonstrates broad potential for transformative applications ranging from next-generation
81 wearables to embodied artificial intelligence.

Results

Architecture of the skin-conformal imaging system

Compared to existing pixel-level stretchable imagers, SkinSight (Fig. 1c) utilizes a hybrid “rigid-in-soft” architecture that shifts the fundamental sensing node from individual pixels to integrated imager modules. In this paradigm, the pixel matrix remains rigid within each imager, while the overall mechanical flexibility of the system is provided by the stretchable serpentine interconnects routing across the soft substrate. Each node is a CMOS sensor with a compact footprint of approximately $1 \times 1 \text{ mm}^2$ and a pixel count of 160 K. In comparison, individual pixels in existing stretchable imagers often span hundreds of micrometers due to fabrication limits, resulting in low total pixel counts (typically $< 1 \text{ K}$). Consequently, each node in SkinSight only occupies a rigid footprint comparable to only a small cluster of these pixels, imposing minimal mechanical constraints on the overall deformable substrate. This compact design elevates the array’s total resolution to megapixel-level and preserving the system’s morphological conformability (Table 2).

This architecture also leads to a highly efficient signal readout strategy. Existing pixel-level stretchable imagers adopt row-column matrix data addressing, demanding a massive number of readout ports (Supplementary Fig. S4). In SkinSight, the CMOS sensor in each node inherently integrates pixel matrix addressing and data serialization, thus requiring only four essential ports (VDD, GND, CLK, and VOUT) per node to continuously transmit 160 K pixel signals. This significantly reduces the overall complexity, allowing the system to be flexibly configured with 1 to over 10 imager nodes to adapt to specific task requirements.

The physical integration of these nodes utilizes a layered structure designed for mechanical robustness (Fig. 1c). In this configuration, a polydimethylsiloxane (PDMS) film serves as the elastic substrate to fully encapsulate the stretchable interconnects. To ensure an unobstructed optical path, a $1 \times 1 \text{ mm}^2$ opening is precisely patterned in the top PDMS layer to expose the CMOS sensor module. To further minimize the node profile, a thin lens is integrated onto each sensor module. These underlying interconnects comprise laser-patterned serpentine copper wires on a polyimide (PI) substrate. This specific geometry, coupled with the PDMS encapsulation, effectively distributes mechanical stress during deformation. As demonstrated in Fig. 1d, this integrated architecture ensures that the optical nodes remain functionally isolated from the strains

of bending, twisting, and stretching. This architecture physically decouples the imaging core from macroscopic strain by encapsulating it within a zero-strain rigid island. Consequently, the intrinsic image quality remains fundamentally unaffected by any external dynamic deformation.

Characterization of the skin-conformal imaging system

The SkinSight system is designed to maintain signal integrity and mechanical conformability under dynamic physical deformation. Figure 2a presents an optical image of a miniaturized imaging module along with its stretchable interconnects. The nano-lens design is highlighted on the left of Fig. 2b, featuring Si_3N_4 nano-pillars fabricated on an SiO_2 substrate. The diameter map of the lens is optimized using a neural nano-optics method¹⁹, spanning approximately 200 μm . To evaluate the optical performance, point spread functions (PSFs) are captured and characterized using red, green, and blue LEDs. A custom display system is utilized to collect paired image data, which are used to train a restoration network for aberration correction and resolution enhancement. (see Methods, Supplementary Fig. S9, S16, and Supplementary Note S2). The flexibility of SkinSight relies on the stretchable serpentine interconnections, which are designed to accommodate the dynamic deformation of the carriers, such as human skin and soft robotic interfaces. Fig. 2c presents the cross-sectional design of the stretchable connections, composed of 100- μm thick PDMS, 18- μm thick copper wires, and a 12.5- μm thick PI substrate. Each copper wire has a width of 0.2 mm, separated by 0.5 mm gaps. This serpentine structure allows the interconnects to stretch, with the overall stretchability tuned by adjusting the arc radius (R) and the straight segment length (L)^{20–22}. The corresponding FEA simulation result under 30% global stretching (matching the typical maximum stretchability of human skin) is presented in Fig. 2d, confirming the strain localization within the soft substrate.

With optical imaging physically decoupled from macroscopic strain by the rigid-in-soft architecture, our characterizations focus on signal transmission reliability. Furthermore, because our encoding-decoding scheme intrinsically tolerates baseline electrical noise, the decoded image fidelity (pixel-level SNR) serves as the definitive metric for validating transmission robustness (see Supplementary Fig. S1).

To experimentally validate the system's signal transmission stability, a single SkinSight node

is evaluated using a custom-built cyclic loading test platform (Fig. 2e). The setup undergoes continuous dynamic deformation across three mechanical states: a compressed state (6.4 cm), the initial flat state (9.2 cm), and a stretched state (11.6 cm). Operating at a loading rate of one cycle per minute and capturing at 30 frames per second (fps), each mechanical cycle consists of 1800 frames. During this dynamic process, the node continuously captures a video stream of a uniform white board (Supplementary Fig. S1). To quantify the SNR, we first extract and analyze the temporal intensity of randomly sampled pixels. Although imaging the uniform white board should yield constant pixel intensities, the relative spatial shifts induced by the macroscopic motion inevitably cause low-frequency baseline drifts in the raw signal. Therefore, a moving average filter with a window size of 200 frames (approx. 1/9 of a cycle) is applied to the temporal sequences (Fig. 2f). This isolates the signal drift (red dashed curve) without distorting its macroscopic envelope, allowing us to strictly extract the true intrinsic noise (the high-frequency residual, green curve) for the final SNR calculation (38.67 dB). Furthermore, fatigue life characterization (Fig. 2g) demonstrates that the interconnects remain stable under repeated mechanical stress. Every 100 cycles, the system pauses at a fixed position to capture 200 static frames for static SNR calculation. The result exhibits a remarkably stable SNR (42.83 dB) even after 500 stretch-compress cycles, confirming the capability of the interconnects to sustain image signal transmission under mechanical fatigue (Supplementary Fig. S1 and Supplementary Movie S1). To further assess the electrical reliability, we continuously monitor the resistance variation of the interconnects throughout the cyclic loading process. The resistance remains within a negligible range (Supplementary Fig. S2 and Movie S2). These combined results conclusively validate the robust performance of SkinSight under repetitive mechanical deformations.

Deformation-adaptive reconstruction

To reconstruct the spatiotemporal environment from the highly deformable SkinSight system, we propose an asynchronous reconstruction paradigm driven by visual geometry foundation models^{23–27} (Fig. 3a). To handle the continuous topological deformations, our paradigm decouples the reconstruction process into an online reconstruction pathway and an asynchronous Gaussian optimization pathway, coupled with a rendering pathway for visual readout. The input undergoes temporal chunking via a sliding window, yielding overlapping chunks. For each chunk, we employ

the $\pi 3$ geometry foundation model²⁸ to back-project 2D pixels into dense 3D point maps, and predict camera view poses without pre-calibrated camera poses. The relative rotation, translation, and scaling between adjacent chunks are analytically solved using a least-squares formulation based on the overlapping poses and point maps. While the online reconstruction pathway rapidly establishes the 3D point map, direct point-based representations and rendering lack fidelity. Therefore, the Gaussian optimization pathway concurrently performs 3D Gaussian splatting optimization in the backend using uniformly sampled frames^{29,30}. We bypass feed-forward Gaussian architectures²⁷, as their current methods lack the robustness required by SkinSight (Supplementary Fig. S8). Lastly, a real-time rendering process transforms these optimized geometric representations (point map or Gaussian) into the final rendered view for perceptual feedback.

As explicitly compared in Fig. 3b, while the online pathway successfully extracts dense and metrically accurate point maps, these point-based representations remain inherently discrete and visually fragmented. As highlighted by the zoomed-in crops, the asynchronous Gaussian optimization effectively addresses this limitation by filling the spatial voids inherent to the raw point map, restoring continuous surfaces and fine textural details. This transformation into a continuous, high-fidelity 3D representation significantly enhances intuitive visual immersion. Furthermore, our approach demonstrates robust superiority over existing baselines (see Supplementary Fig. S6, S7, and S8). Conventional pipelines like COLMAP^{31,32} are computationally slow and typically yield sparse, noisy point clouds. While recent geometry models like InstantSplat³³ also utilize geometric foundation models for one-time global reconstruction, they function as static global estimators that lack the capacity for online updates, rendering them structurally unsuited for continuous video streams.

Since our method relies purely on visual streams from the uncalibrated array, the reconstructed geometry inherently contains a scale ambiguity, capturing only relative dimensions. To achieve absolute metric recovery, a universal physical reference is required. This metric scale factor can be obtained using auxiliary depth sensors, IMUs, or an object with known physical dimensions (see Methods). To quantitatively evaluate the dimensional accuracy, we reconstructed a complex desktop scene comprising various household objects (Fig. 3c). By anchoring the metric scale to the length of the red rectangular box, we extracted the physical dimensions of the remaining objects utilizing 3D bounding boxes from the reconstructed point maps. Compared with

the ground-truth measurements (Supplementary Fig. S3), our reconstruction achieves an exceptionally low average dimensional error of 1.52% (Table 1).

Crucially, this asynchronous paradigm is intrinsically capable of reconstructing dynamic environments containing moving targets. The online reconstruction pathway can dynamically prune obsolete chunks and recursively integrate newly observed chunks, ensuring the temporal coherence of the reconstructed scene. To rigorously validate this capability, we conducted an experiment tracking a target (a sunflower ornament) translating on a motorized conveyor belt (Fig. 3d and Supplementary Movie S3). Specifically, we tracked the target within the reconstructed scene to pinpoint its absolute 3D positions³⁴. By applying a least-squares linear regression to the 3D spatial displacements over time, we directly quantified the target’s physical velocity. The visually derived average velocity is 1.669 cm/s, which exhibits a remarkable consistency with the programmed mechanical speed of 1.666 cm/s, achieving a highly accurate dynamic measurement.

Augmented vision via skin-conformal system integration

The SkinSight system establishes a versatile visual interface for human vision augmentation, transforming deformable surfaces into active sensory extensions. We demonstrated this paradigm by integrating the SkinSight onto a human palm to navigate narrow, occluded environments (Fig. 4a). In this scenario, both the human eye and the rigid headset camera are physically blocked by the opaque box, permitting only a severely restricted view of the internal space through a narrow window. Conversely, the flexible SkinSight-enabled hand can be inserted into the box through the narrow window, bypassing these physical obstructions to comprehensively capture the internal scene. Based on the captured videos from the inserted SkinSight, we successfully reconstruct high-quality 3D geometric entities, including dense point maps, Gaussian representations, and mesh models of the occluded internal scene (Fig. 4c).

To grant the user a functional see-through perception, we implement spatial registration between the SkinSight coordinates and the headset coordinate system, enabling the dynamic overlay of reconstructed entities directly onto the headset’s camera view (Fig. 4b). The resulting AR visualization explicitly validates the performance of our dual-pathway architecture. As demonstrated in the top row of Fig. 4d, the registered 3D point maps provide fast, geometrically

accurate feedback that effectively localizes hidden objects within the user’s field-of-view. However, as further highlighted by the zoomed-in crops, these point-based representations suffer from visual fragmentation and blurring artifacts. The asynchronous Gaussian optimization (bottom row) directly addresses this by filling spatial gaps and restoring photorealistic surface textures. This transition from discrete points to continuous, high-fidelity Gaussian representations is essential for creating a naturalistic sense of visual immersion, ensuring that the augmented overlay is not only metrically precise but also perceptually intuitive. Crucially, both pathways achieve accurate spatial alignment with the physical objects (e.g., the red box) partially visible through the narrow window, confirming the reliability of our cross-platform coordinate registration. Supplementary Movie S4 demonstrates this functional see-through capability through a complete pipeline of coordinate alignment, reconstruction, and rendering.

Beyond human augmentation, SkinSight demonstrates significant utility in soft robotics, where rigid visual sensors are often mechanically incompatible. When integrated onto a soft gripper (Supplementary Fig. S11, S17 and S18), the conformal SkinSight array maintains robust, blind-spot-free visual sensing, whereas a commercial RGB-depth camera suffers from severe occlusions. This comparison demonstrates that our flexible system offers significant advantages in constrained environments. In summary, by combining an ultra-flexible hardware design with AI-empowered reconstruction, SkinSight successfully transforms highly deformable surfaces into intelligent sensory interfaces. This versatile platform provides unprecedented visual perception capabilities for next-generation human-computer interaction and embodied artificial intelligence.

Discussion

We introduce SkinSight, a skin-conformal artificial vision system that combines high pixel resolution with mechanical flexibility. Unlike existing flexible vision systems that suffer from low total pixel counts, SkinSight achieves a substantial leap in resolution through a hybrid rigid-in-soft architecture. This massive increase in total pixel number (160 K pixels per node) enables the capture of high-resolution visual data necessary for advanced machine vision tasks. Furthermore, compared to existing unstructured camera arrays, SkinSight revolutionizes the physical form factor of visual sensors. By transitioning from bulky hardware to thin and wearable formats, the system empowers dynamic hosts, such as human hands, into sensory extensions to perform high-dimensional visual perception in severely restricted environments.

The practical deployment of SkinSight hinges on three key engineering characteristics: integration limits, fabrication yield, and failure modes. First, regarding integration limits, the system achieves good conformability through the moderate stretchability of the serpentine interconnects between the rigid sensor islands. This design prioritizes high pixel counts and massive data throughput while maintaining sufficient flexibility for dynamic hosts, offering a highly practical balance for advanced visual tasks. Second, in terms of fabrication yield, SkinSight leverages mature industrial processes including flexible printed circuits (FPC) and automated surface-mount technology (SMT). This approach ensures highly reproducible fabrication, yielding robust electrical connectivity and reliable encapsulation without the inconsistency of manual lab-scale assembly. Third, concerning failure modes, the ultra-miniature size of the imagers inherently restricts the soldering area at the rigid-flex interface, making these connection points the primary mechanical weak link compared to the highly flexible serpentine wires. Our current encapsulation strategy effectively protects these connections to ensure stable data transmission. Crucially, as validated by our cyclic loading evaluations, the system maintains intact functionality without any observable solder joint delamination even after hundreds of stretch cycles (Fig. 2). Future optimizations of these specific interfaces will further enhance the system’s long-term resilience under extreme mechanical stress.

As summarized in Table 2, SkinSight combines the high-resolution capabilities of unstructured camera arrays with the mechanical flexibility of intrinsically flexible artificial vision. Unstructured camera arrays rely on bulky data readout interfaces and rigid assemblies, limiting

their mechanical conformability. Conversely, intrinsically flexible vision systems excel in extreme deformation but suffer from severe resolution bottlenecks. This limitation stems from both fabrication and data readout constraints. Lab-scale micro-transfer printing techniques struggle with the yield and scalability required for massive pixel integration. Because these flexible systems typically employ direct row-column matrix addressing, scaling up the pixel count leads to a significant growth in the number of external wires for data readout (Supplementary Fig. S4). SkinSight circumvents these physical constraints through its rigid-in-soft architecture, by leveraging commercial CMOS sensors for in-node matrix addressing, which enables serial data transmission via 4-wire stretchable interconnects. Because each node operates independently without complex cross-wiring, the total node number of SkinSight is not physically restricted by the readout architecture, making the system modularly scalable to meet various application demands (Supplementary Fig. S5). Functionally, it aligns with unstructured camera arrays by delivering the megapixel-level data required for advanced vision applications. Mechanically, it aligns with flexible bio-integrated electronics.

At the system level, the massive data stream generated by the highly deformable array presents a fundamental computational challenge. To address this, SkinSight leverages recent breakthroughs in geometry foundation models for robust 3D reconstruction and Gaussian splatting for high-fidelity rendering feedback. These AI-driven advancements are crucial for extracting meaningful spatial information from the complex, unanchored data flows of our flexible sensors. As demonstrated by our human hand integration, this technology effectively transforms biological limbs into active sensory extensions, enabling users to comprehensively perceive and navigate environments that are visually inaccessible.

In summary, the flexibility of SkinSight is not merely a physical feature but a transformative enabler. By combining an ultra-flexible array-based imaging system with AI-empowered reconstruction, SkinSight successfully transforms highly deformable surfaces into intelligent visual interfaces. This evolution from passive measurement tools into active and conformal sensory systems establishes a robust foundation for the next generation of human-computer interaction and embodied artificial intelligence.

Methods

Stretchable interconnects fabrication and system integration. The fabrication and integration process involves three primary steps: designing and manufacturing the serpentine stretchable interconnects, soldering the imagers, and encapsulating the system with PDMS. The serpentine interconnects are fabricated using mature flexible printed circuit (FPC) technology. This layout comprises an 18- μm copper layer for power and signal transmission, patterned on a 12.5- μm polyimide substrate. Subsequently, the miniaturized imagers (OV6946, OmniVision Technologies) are soldered onto the designated contact pads at the interconnect terminals. Prior to PDMS bonding, a cavity is punched into the top PDMS film to accommodate the imager housing. The surfaces of both the serpentine interconnects and the PDMS films undergo oxygen plasma treatment to enhance interfacial adhesion. The top and bottom PDMS layers are then thermally bonded to the flexible circuit under pressure and cured at 80 °C for 6 hours. Finally, additional PDMS is cast around the imager base to reinforce local structural integrity (Supplementary Fig. S14). The image data from the miniaturized sensors is transmitted through the serpentine interconnects and processed by ADC chips (OV426, OmniVision Technologies).

The primary manufacturing bottleneck of the system involves reliably soldering the ultra-miniature imagers at the rigid-flex interface. This challenge is successfully overcome by designing customized alignment fixtures and employing precision automated SMT reflow soldering to standardize the connections. Beyond this critical integration step, the remaining fabrication procedures are highly straightforward and reproducible. The stretchable interconnects leverage mature industrial FPC processes to provide a highly reliable electrical baseline, and the final PDMS encapsulation is completed through a simple casting and bonding process. Consequently, the entire system assembly achieves exceptional reproducibility and a high overall yield without demanding highly complex laboratory techniques.

Electromechanical cyclic loading and optoelectronic stability characterization. The dynamic mechanical evaluations detailed in the Results section are conducted using a custom-built motorized linear stage platform (Fig. 2, Supplementary Fig. S1). The platform is driven by a programmable stepper motor to ensure precise control over the applied strain and continuous displacement, operating at a rate of approximately one complete stretch-release-compress cycle per minute. For the dynamic signal-to-noise ratio (SNR) quantification, the SkinSight node

captures the uniform white target under constant illumination at a frame rate of 30 frames per second (fps).

To isolate the intrinsic sensor noise from the macroscopic motion artifacts, data processing is performed using custom analytical scripts. The raw temporal intensity vector for each pixel is first extracted. The low-frequency baseline drift, which arises from the relative translation between the sensor and the target surface, is mathematically estimated by convolving the raw signal with a 200-frame moving average filter. This calculated baseline is then subtracted from the raw data to strictly isolate the high-frequency residual noise. The temporal SNR is subsequently computed as the ratio of the mean raw signal intensity to the standard deviation of this high-frequency residual noise.

For the fatigue life characterization, the automated protocol programs the linear stage to execute 500 continuous stretch-compress cycles. The system is designed to automatically pause the motion every 100 cycles, holding the substrate at the flat state to capture 200 static frames. Since the substrate remains completely stationary during these captures, motion-induced baseline drift is inherently eliminated. Consequently, the static SNR is directly computed using the raw mean and standard deviation of these frames without any moving average filtering.

Nano-lens design. The relationship between the nano-pillar diameter and its modulation phase (structure-to-phase mapping) is fitted using a differentiable polynomial:

$$\varphi = 2\pi \times (6.051 - 0.0203\lambda + 2.26\eta + (1.371 \times 10^{-5})\lambda^2 - 0.00295\lambda\eta + 0.797\eta^2), \quad (1)$$

where λ is the wavelength of the light in nano-meters, η is the duty of the nano-pillar, and φ is the modulation phase. The duty is defined as: d_p/d_u , where d_p is the diameter of the nano-pillar, and $d_u = 350$ nm is the size of the square unit. The relationship between pillar diameter and phase modulation was computed using the finite-difference time-domain (FDTD) method (RCWA Solver in Lumerical, ANSYS Inc). Supplementary Fig. S12 visualizes the FDTD results and polynomial fitting results. The height of the pillar is 705 nm, and the diameter of the whole nano-lens is 200 μ m, approximately comprising 571 $\times$ 571 units. The structural map of the lens is optimized by employing the neural nano-optics framework developed in the PyTorch environment. For fabrication, we begin with a 200- μ m polished fused silica wafer, deposit 705 nm of silicon nitride (Si_3N_4), and exposure with an electron-beam lithography system to obtain the nano-pillar structure.

Image restoration and semantic segmentation. For image restoration, we leverage established neural network architectures to accommodate specific application requirements. A lightweight U-Net is sufficient to provide fast inference for recovering high-quality, aberration-free images. Furthermore, a diffusion-based generative network^{35–37} can be optionally incorporated to further elevate the reconstruction quality. LoRA-based fine-tuning³⁸ is adopted for parameter-efficient adaptation. The optimization process minimizes a joint objective comprising Mean Squared Error (MSE) and Learned Perceptual Image Patch Similarity (LPIPS)³⁹ losses, ensuring both pixel-wise metric accuracy and perceptual visual quality. Finally, for downstream scene understanding, semantic segmentation is performed utilizing the mature Grounded-DINO⁴⁰ framework in a zero-shot manner, requiring no additional training or modifications. Further algorithmic details and extended evaluations are provided in Supplementary Note 2.

Asynchronous 3D reconstruction and Gaussian optimization. To handle continuous topological deformations without pre-calibrated camera extrinsics, the spatial reconstruction is decoupled into an online geometric reconstruction pathway and an asynchronous Gaussian optimization pathway. The continuous multi-view video stream undergoes temporal chunking via a sliding window to yield overlapping frame chunks C_k . Each chunk $C_k = \{\mathbf{I}_1, \mathbf{I}_2, \dots, \mathbf{I}_L\}$ comprises a sequence of L images, which is processed as a collective input by the $\pi 3$ visual geometry foundation model $\Phi_{\pi 3}$. The functional mapping of the model is defined as:

$$\Phi_{\pi 3}(C_k) \rightarrow \{\mathbf{P}_t, \mathbf{c}_t, \Theta_t\}_{t=1}^L, \quad (2)$$

where $\mathbf{P}_t \in \mathbb{R}^{H \times W \times 3}$ represents the reconstructed dense 3D point map, $\mathbf{c}_t \in \mathbb{R}^{H \times W}$ denotes the per-pixel confidence score, and $\Theta_t \in \text{SE}(3)$ signifies the locally consistent camera pose for each frame t within the chunk. By implicitly encoding cross-view semantic and geometric features, the model directly regresses these spatial representations in a local coordinate space without requiring explicit feature matching or camera calibration.

The relative transformation between adjacent chunks C_k and C_{k+1} can then be efficiently analytically solved using a least-squares formulation. Given the set of corresponding 3D points $\{\mathbf{P}_k\}$ and $\{\mathbf{P}_{k+1}\}$ derived from the overlapping point maps, we minimize the weighted registration error:

$$E = \sum c_k^i \rho(\|\mathbf{P}_k^i - (s\mathbf{R}\mathbf{P}_{k+1}^i + \mathbf{T})\|_2), \quad (3)$$

where $\mathbf{R}$, $\mathbf{T}$, and s represent the rotation matrix, translation vector, and scaling factor, respectively. c_k^i denotes the confidence value for the i -th point correspondence extracted from the reference chunk C_k . This approach ensures that the global coordinate alignment is primarily driven by high-confidence scene structures, with the confidence map effectively suppressing the influence of dynamic objects. Meanwhile, the Huber loss $\rho(\cdot)$ is utilized to handle the geometric outliers. This sequential alignment integrates all local chunks into a unified global coordinate system.

In our implementation, we set the chunk length 30 with an adjacent overlap of 15 frames. The geometry foundation model requires approximately 2.0 s per chunk, while the spatial alignment between C_k and C_{k+1} takes about 2.2 s. Performance benchmarks were conducted on a PC equipped with an NVIDIA GeForce RTX 4090 GPU and an Intel Core i9-13900K CPU. It is worth noting that the current processing speed was achieved without extensive low-level engineering optimizations, suggesting significant potential for further throughput enhancement and real-time deployment. For example, the point map reconstruction and alignment can be executed in parallel via an asynchronous dual-thread architecture to further minimize reconstruction latency. Notably, while the reconstruction operates asynchronously on a chunk-by-chunk basis, this does not impede the real-time rendering process, which maintains a continuous visual output while the displayed scene content is updated incrementally upon the completion of each chunk.

The registered dense point maps from the online reconstruction pathway provide the geometric and color priors to the Gaussian optimization, where the environment is modeled as a collection of 3D Gaussian primitives defined by their mean positions $\boldsymbol{\mu}$, covariance matrices $\boldsymbol{\Sigma}$, spherical harmonic (SH) coefficients $\mathbf{h}$ for view-dependent color, and opacity α . The Gaussian reconstruction enables photorealistic visual renderings by projecting these 3D primitives into 2D image space via differentiable α -blending. The Gaussian primitive attributes are iteratively optimized via gradient descent by minimizing a joint loss function:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{color} + \lambda_2 \mathcal{L}_{depth} + \lambda_3 \mathcal{L}_{normal}, \quad (4)$$

$$\mathcal{L}_{color} = \sum_k \sum_i \|\hat{\mathbf{I}}_k^i - \mathbf{I}_k^i\|_2^2, \quad (5)$$

$$\mathcal{L}_{depth} = \sum_k \sum_i \|\hat{\mathbf{D}}_k^i - \mathbf{D}_k^i\|_1, \quad (6)$$

$$\mathcal{L}_{normal} = \sum_k \sum_i |\nabla \mathbf{I}^i| \|\hat{\mathbf{N}}_k^i - \mathbf{N}_k^i\|_1, \quad (7)$$

Where k is the index of the sampled images, i denotes the pixel index in the image. $\mathbf{I}_k$ is the input images, $\mathbf{D}_k$ and $\mathbf{N}_k$ denote the input depth map and normal map computed from the registered point map $\mathbf{P}_k$. $\hat{\mathbf{I}}_k$, $\hat{\mathbf{D}}_k$, and $\hat{\mathbf{N}}_k$ represent the rendered images, depth maps, and normal maps. Benefiting from the dense point map as a geometric initialization, the Gaussian optimization converges rapidly in just 1,000 iterations, requiring approximately 10 s. Following optimization, the geometry mesh can be optionally reconstructed via the TSDF integration⁴¹ and the marching cubes algorithm⁴². Refer to Supplementary Note 1 for more details. Direct feed-forward Gaussian architectures are intentionally bypassed in this pipeline, as current models often struggle to generalize across such complex scenes captured by the deformable system (SkinSight).

Metric scale recovery and dynamic target tracking. Since the reconstruction relies solely on visual streams from an uncalibrated array, the generated 3D geometry inherently contains a scale ambiguity. Absolute metric recovery is achieved by introducing a universal physical reference with a known dimension into the scene. The global metric scale factor s_{metric} is determined by $s_{metric} = L_r/L_p$, where L_r and L_p represent the real physical dimension and the measured dimension in the reconstructed point map, respectively. This factor allows for the extraction of absolute physical dimensions for the remaining reconstructed objects.

For dynamic targets, the online reconstruction pathway recursively integrates newly observed chunks while pruning obsolete ones to maintain temporal coherence. The spatial positions $X(t)$ are estimated by employing an object tracker³⁴ on the rendered images and mapping the tracked 2D coordinates to the 3D space. The physical velocity V of the moving target is then quantified by applying a least-squares linear regression to these spatial displacements over time: $X(t) = Vt + X_0$. The velocity is directly derived from the optimized slope of this linear model.

Data availability

The datasets generated in this study have been deposited in the Zenodo database under accession code 10.5281/zenodo.20049665⁴³ [<https://doi.org/10.5281/zenodo.20049665>].

456 **Code availability**

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Contributions

L.F. initiated and supervised this study. L.F. and X.Y. developed the original idea. X.Y. and J.Z. conceived the research and methods, and performed the experiments. X.Y. and Y.L. built the system. X.Y. and Y.W. designed, fabricated, and tested the nano-lenses. Y.W. fabricated the miniaturized imagers. X.Y., J.Z., and L.F. wrote the manuscript. All authors discussed the research.

Competing interests

The authors declare no competing interests.

Table 1: Dimensional accuracy of physical objects in the reconstructed scene. The length of the Top red box is used as the metric reference. The length of the Top red box (highlighted in bold) is used as the metric reference to anchor the global metric scale.

Object	Measured physical dimension (cm)	Reconstructed physical dimension (cm)	Error
Top red box (reference)	18.0 x 7.0 x 13.0	18.0 x 7.1 x 13.2	1.48%
Bottom red box	18.0 x 7.0 x 13.0	18.2 x 7.0 x 13.1	0.63%
Tall green cup	7.6 x 7.6 x 14.5	7.5 x 7.5 x 14.3	1.34%
Pepsi can	6.5 x 6.5 x 11.5	6.4 x 6.4 x 11.6	1.32%
Sunflower ornament	7.8 x 5.5 x 14.3	7.9 x 5.3 x 14.3	1.64%
White glasses case	16.2 x 5.5 x 2.4	16.2 x 5.4 x 2.6	3.38%
Infrared thermometer	15.2 x 7.0 x 3.5	15.1 x 7.1 x 3.5	0.70%
Computer mouse	6.4 x 10.8 x 3.4	6.3 x 10.75 x 3.5	1.66%
average	/	/	1.52%

Table 2: Comparison of our skin-conformal rigid-in-soft array-based imaging system with representative flexible artificial vision systems and unstructured camera array.

	System	Size	Flexibility during imaging	Fabrication complexity	Data readout	Node size	Pixels per node	Node number
Flexible artificial vision system	Hemispherical eye camera ⁷	Thin and small	Fixed	Lab-scale transfer printing	Direct row-column matrix addressing (Requires massive stretchable wire bundles)	500 μm	1	16 $\times$ 16
	Dynamically hemispherical eye camera ⁸		Focus tunable			500 μm	1	16 $\times$ 16
	Arthropod-inspired eye camera ⁶		Fixed			160 μm	1	16 $\times$ 16
	Kirigami-design imager ¹²		Fixed			300 μm	1	32 $\times$ 32
	Anti-distortion bioinspired camera ⁹		Fixed			$\sim$ 87 μm	1	128 $\times$ 32
Unstructured camera array	Unstructured camera arrays (machine vision) ²	Large	Fixed	Assembly of commercial cameras	In-node matrix addressing + Bulky cables or offline local storage (e.g., MIPI/USB wires)	$\sim$ 30 mm	2048 $\times$ 2048	14
	Unstructured camera arrays (GoPro) ²		Fixed			$\sim$ 30 mm	1920 $\times$ 1080	5
	Flexible Camera Arrays ₃		Bendable			$\sim$ 30 mm	640 $\times$ 480	20
	Modular hierarchical array camera ^{4,5}		Rotatable			$\sim$ 30 mm	4000 $\times$ 3000	14
Skin-conformal rigid-in-soft array-based imaging system (Ours)	SkinSight	Thin and small	Stretchable, bendable	FPC + SMT	In-node matrix addressing + Stretchable 4-wire interface	1,000 μm	400 $\times$ 400	Scalable

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592 **Fig. 1: A skin-conformal rigid-in-soft array-based imaging system.** **a,b,** Architectural
593 paradigm shift from a conventional rigid camera array to a skin-conformal rigid-in-soft array-
594 based imaging system. Through decoupling, miniaturizing, and conforming, the proposed system
595 overcomes the spatial constraints of conventional camera arrays, enabling visual integration on
596 complex deformable carriers, such as human skin and soft robots. **c,** Schematic of the rigid-in-soft
597 architecture. The system consists of distributed miniaturized imager modules linked by stretchable
598 interconnections on an elastic substrate. The exploded view details the component layers of a
599 single node: protective polydimethylsiloxane (PDMS) encapsulation, polyimide (PI)-supported
600 copper wiring for signal transmission, a complementary metal-oxide-semiconductor (CMOS)
601 sensor, and a thin lens. **d,** Optical images showing the mechanical compliance of the imaging
602 system. It can be bent, twisted, and stretched.

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Fig. 2: Design and characterization of the skin-conformal rigid-in-soft array-based imaging system. **a**, Conceptual schematic and optical photographs of the hybrid rigid-in-soft architecture, showcasing the distributed imager modules and stretchable serpentine interconnects. **b**, Design of the nano-lens, utilized to minimize the overall node thickness. **c**, Cross-sectional layout and geometric parameters of the serpentine interconnects, comprising a polyimide (PI) substrate, copper traces, and polydimethylsiloxane (PDMS) encapsulation. **d**, Finite element analysis (FEA) illustrating the strain distribution of the serpentine interconnects under 30% stretching. **e**, Cycle loading test platform. Images of the testing setup during dynamic cyclic deformation, showing three states: compressed (6.4 cm), original flat (9.2 cm), and stretched (11.6 cm). **f**, Signal integrity under cyclic deformation. A moving average filter (window size = 200, approx. 1/9 of the 1800-frame mechanical cycle) is applied to accurately track the motion-induced low-frequency baseline drift and isolate the high-frequency temporal noise, yielding a robust operational signal-to-noise ratio (SNR) of 38.67 dB. **g**, Fatigue life characterization, demonstrating highly stable signal transmission over 500 stretch-compress cycles.

Fig. 3: Deformation-adaptive reconstruction. **a**, Schematic of the asynchronous computational architecture. An online reconstruction pathway processes the temporally chunked video stream to extract view poses and three-dimensional (3D) point maps. Concurrently, an asynchronous Gaussian optimization pathway refines the scene for high-fidelity rendering. **b**, Comparative analysis of 3D point map and Gaussian representation. Although the registered 3D point map (left) inferred by the geometry foundation model is already dense, it still results in a fragmented visual experience. The Gaussian pathway transforms this initial estimate into a continuous, photorealistic 3D representation (right), effectively bridging metric recovery with visual immersion. **c**, Quantitative evaluation of metric geometric accuracy. The physical dimensions of diverse objects (bounding boxes) in a complex desktop scene are extracted and compared against ruler measured ground truth (Table 1). **d**, Dynamic scene reconstruction. The system recursively updates the representation to track a moving sunflower ornament on a motorized conveyor belt. The panels display the reconstructed 3D point maps, tracked camera poses (red pyramids), and corresponding captured images at two distinct time points ($t = 0.5$ s and $t = 29.6$ s). **e**, Quantitative evaluation of the target’s velocity. By applying a least-squares linear regression to the 3D spatial displacement over time, the visually derived velocity matches the programmed mechanical speed.

Fig. 4: Augmented vision via integration of the skin-conformal array-based imaging system.

a, Demonstration of the augmented vision paradigm via the system’s integration. Snapshots extracted at discrete time points t from both the augmented reality (AR) glasses and the conformal array’s video streams illustrate the exploration of occluded objects within an opaque box. **b,** Coordinate alignment and spatial registration. The continuous poses of the system’s cameras and the AR glasses camera are plotted by the color-coded path (colorbar indicates temporal progression). By matching co-visible visual features from the exterior environment, the system’s coordinates are registered to the AR glasses coordinate system, enabling accurate overlay of reconstructed geometry. **c,** Reconstructed 3D point map, Gaussian representation, and mesh model. **d,** Representative AR overlay results at two distinct time points ($t = 51$ s and $t = 81$ s). Point map rendering (top) provides fast geometric feedback, while Gaussian rendering (bottom) offers high-fidelity visual immersion, as explicitly highlighted in zoomed-in crops. The rightmost crops demonstrate accurate spatial alignment between the overlay and the physical object (red box) partially visible through the narrow window.